

SATQUERY AI — ANALYSIS REPORT

Interactive Agentic Vision-Language Remote-Sensing Intelligence Platform

Execution ID:	exec_4c44c0b0f6
Timestamp:	2026-08-28T18:37:50Z
Analysis Mode:	OPTICAL_SAR
Identified Task:	OPTICAL_SAR_FUSION
User Query:	"Perform cross-modal optical and SAR fusion."
Overall Confidence:	60.0% (model_derived)

AI Intelligence Analysis Result

Multimodal Optical+SAR Cross-Modal Fusion completed. Combined Sentinel-2 optical multispectral features with Sentinel-1 SAR radar backscatter representations (Cross-modal feature correlation: 0.44).

Observable Agent Execution Trace

Step Name	Status	Detail
Input Validation & Geospatial Inspection	COMPLETED	Mode: OPTICAL_SAR Primary: img_sample_opt_01_S2A_MSIL2A_20170613T101031_N9999_R022_T33UUP_45_68.png (120x120 PNG)
Query Intent & Requirement Classifier	COMPLETED	Intent: OPTICAL_SAR_FUSION Target: multimodal_land_cover
Specialist Model Routing & Inference	COMPLETED	Model Loading ✓ Checkpoint: ResNet18_Multimodal_OpticalSAR_PyTorch Device: CPU Mode: REAL_MODEL Inference ✓
Evidence Packaging & Dynamic Confidence Calibration	COMPLETED	Confidence: 60% (model_derived) Evidence: 3 items

Generated Visual Evidence

Title	Type	Reference
Primary Satellite Input	image	/storage/uploads/img_sample_opt_01_S2A_MSIL2A_20170613T101031_N9999_R022_T33UUP_45_68.png
Secondary Input Image	image	/storage/uploads/img_sample_sar_01_S1B_IW_GRDH_1SDV_20170612T165809_33UUP_45_68_S1_VV.png

Optical + SAR PyTorch Fused Representation	optical_sar_overlay	/storage/generated/real_neural_fused_267208a4.png
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